

trolled LCD module (LCD1) for display. Typically, a 16-character, 2-line LCD module has eight data lines (D0 through D7) for data transfer. In this project only four data lines (D4 through D7) are used for data transfer. Data lines D0 through D3 are disabled. Data transfer between HD44780 and the microcontroller completes after the 4-bit data is transferred twice. Port-D pins PD0 through PD3 of the microcontroller (IC2) are connected to data lines D4 through D7 of LCD1. LCD1 control lines, i.e., read/write (R/W), register-select (RS) and enable (EN), are connected to PD5, PD4 and PD6 of IC2, respectively.

Keypad. A total of 16 pushbutton switches are used and arranged in a 4x4 keypad matrix on the PCB. One terminal of each pushbutton is connected to common ground, while the other terminal is connected to resistors R1 through R16 in series. Refer Fig. 2 and Table I for details.

In this project, we use the voltage divider circuit principle for identification of each key/switch pressed. Here, 1-kilo-ohm resistor R19 is fixed and connected to 5V power supply as shown in Fig. 2. When you press any pushbutton switch, the corresponding resistor, which is connected in series with the pushbutton, comes in picture to form a voltage divider circuit. As a result, you get a variable voltage at pin 39 (PA1) of IC2, which is analogue-to-digital converter channel-1 pin or ADC1 pin of the microcontroller.

Voltage obtained at ADC1 pin is given by the relationship:

$$V_{out} = V_{in} \times \{R_i / (R_{19} + R_i)\}$$

R19 has a constant value of one kilo-ohm and V_{in} is also a constant value of 5V. R_i value is decided when you press any of the pushbutton keys from the keypad. It can be any of the resistors R1 through whose values are listed in Table I.

Software program

To work with Atmel AVR microcontroller using 'C' programming lan-

TABLE I
RESISTORS AND
CORRESPONDING
ADC VALUES

Resistor	Value in ohms	Expected at PA1/ADC1 (Volt)	ADC count value
R1	100	0.455	93
R2	200	0.833	171
R3	300	1.154	236
R4	400	1.429	293
R5	500	1.667	341
R6	600	1.875	394
R7	700	2.059	422
R8	800	2.222	455
R9	900	2.368	485
R10	1000	2.500	512
R11	1100	2.619	536
R12	1200	2.727	559
R13	1300	2.826	579
R14	1400	2.917	597
R15	1500	3.000	614
R16	1600	3.077	630

guage, you need two software tools: AVR Studio and WinAVR. Both of these tools are freely available for download from the Internet. The program for the microcontroller is written in 'C' language using AVR Studio 4.

'C' program functions. Functions are written for the ADC

module, LCD1 and keypad using 'C' program. Working is very simple as all pushbuttons or switches are connected to only one pin of the microcontroller (ADC1). The program reads the input switches through ADC1 pin periodically. After decoding the ADC value, the corresponding switch number/value is displayed on LCD1. Hysteresis approach is used for each pushbutton to analyse the ADC value, and based on that value, the program detects/decodes the pushbutton being pressed. The decoding is done using following functions:

```
Display_Key_on_LCD (unsigned long ADC_value)
```

```
// This function prints the key pressed on the LCD after decoding the ADC value
```

```
DisplayKeyPressed()
```

```
// This function executes every 100 msec
```

```
ReadKey()
```

```
// This function is used for acquiring ADC value on ADC1 pin
```

Compiling and programming.

AVR Studio is an integrated development environment that includes an editor, assembler, etc. WinAVR is a GCC-based compiler for AVR. It also includes a program called Programmer's Notepad that can be used to edit and compile 'C' programs. WinAVR setup file is available at <http://winavr.sourceforge.net/>

Remember to install AVR Studio before installing WinAVR. First, launch AVR Studio from the desktop. You need to create a new project for the main.c code. Next, click 'Build→Rebuild All' to compile the 'C' code.

If there is no error message, a file called Single_wire.hex will be generated. This file contains the machine code that is ready to be programmed into the ATmega32 microcontroller. The file is stored in sub-folder '\default' of your project.

If there are error messages, check your 'C' code. Most often these are caused by some typo or syntax errors.

PARTS LIST

Semiconductors:	
IC1	- 7805, 5V voltage regulator
IC2	- ATmega32 microcontroller
BR1	- 1A bridge rectifier
LED1	- 5mm LED
Resistors (all 1/4-watt, 5% carbon):	
R1-R16	- Refer Table I
R17	- 180-ohm
R18, R19	- 1-kilo-ohm
R20	- 220-ohm
VR1	- 1-kilo-ohm preset
Capacitors:	
C1	- 1000µF, 35V electrolytic
C2	- 10µF, 16V electrolytic
C3	- 120pF ceramic disk
Miscellaneous:	
CON1	- 2-pin terminal connector
LCD1	- 16x2 LCD module HD44780
S1-S16	- Pushbutton switch
X1	- On/off switch
X1	- 230V AC primary to 9V, 250mA secondary transformer
	- 230V AC mains power supply